

Project Proposal

Group 30

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1 Overview

This project focuses on segmenting adult gliomas from MRI scans using a 2D U-Net convolutional neural network. The BraTS 2021 dataset provides four co-registered MRI sequences for each subject: T1, T1CE, T2, and T2-FLAIR, together with an expert-annotated 3D tumor segmentation mask. Our model will operate on 2D axial slices extracted from these volumes. For each slice, the corresponding images from all four MRI sequences are stacked as input channels, and the model predicts which image regions correspond to background or to one of the tumor sub-regions defined by BraTS.

Manual tumor segmentation requires trained raters to delineate tumor regions throughout a 3D MRI volume and is therefore time-consuming. An automated model can generate an initial segmentation that can be reviewed and corrected by an expert, reducing the amount of manual delineation required. Automated segmentation is also useful for quantitative research because a fixed model applies the same learned segmentation procedure to every scan, providing more repeatable measurements across large datasets than repeated manual segmentation. This can make it easier to compare tumor regions across subjects or across scans collected at different time points.

Brain tumor segmentation is challenging because gliomas vary substantially between subjects in size, shape, location, and appearance. Tumor tissue can also resemble surrounding healthy tissue, and tumor regions may occupy only a small fraction of an MRI slice, creating a strong imbalance between tumor and background. In addition, different tumor sub-regions are distinguished by patterns across T1, T1CE, T2, and T2-FLAIR rather than by a single MRI sequence alone. The model must therefore combine information from the four aligned inputs while identifying irregular tumor boundaries and separating tissue classes with similar appearances.

2 Task Definition

2.1 Machine Learning Task

The project is a supervised semantic segmentation task. Rather than assigning one class to an entire MRI slice, the model performs dense multiclass classification by assigning a class to each spatial location in the slice. The ground-truth labels are provided by the expert-annotated segmentation masks included with the BraTS 2021 dataset. Each location is assigned exactly one class, making this a single-label, four-class segmentation problem rather than a multi-label classification problem.

2.2 Model Input

Each model input consists of four corresponding axial MRI slices from the same subject and anatomical position. The slices are taken from the T1, T1CE, T2, and T2-FLAIR volumes, which have already been co-registered so that the same spatial location corresponds to the same anatomy in each sequence. The four 240×240 slices are stacked as separate input channels, producing a $240 \times 240 \times 4$ input to the 2D U-Net.

Using all four sequences allows the model to use differences in tissue appearance across the MRI sequences when distinguishing tumor sub-regions. For example, T1CE provides information about contrast-enhancing tissue, while T2 and T2-FLAIR provide different contrast for tissue with increased water content, including regions associated with edema.

2.3 Model Output and Segmentation Classes

For each input slice, the model produces a 240×240 segmentation map with one predicted class for each spatial location. The original BraTS 2021 segmentation masks use labels 0, 1, 2, and 4. Before training, label 4 will be remapped to 3 so that the model predicts four consecutive classes:

- Class 0: background / non-tumor tissue
- Class 1: necrotic and non-enhancing tumor core (NCR/NET)
- Class 2: peritumoral edema / invaded tissue (ED)
- Class 3: enhancing tumor (ET), corresponding to label 4 in the original BraTS masks

The predicted 2D segmentation therefore corresponds directly to the labels from the matching axial slice of the original 3D BraTS segmentation mask.

2.4 Technical Challenges

The segmentation task is difficult because the four classes are highly imbalanced: background tissue occupies most of an MRI slice, while individual tumor sub-regions may contain relatively few pixels. Gliomas also vary substantially between subjects in their size, shape, location, and appearance, so the model must generalize across tumors with very different morphologies. Boundaries between tumor sub-regions may be irregular or visually ambiguous, requiring the model to distinguish tissue classes using

information distributed across the four MRI sequences. Finally, because the proposed model operates on individual 2D slices, it does not directly use information from adjacent slices in the 3D MRI volume.

3 Solution

U-Net is a convolutional neural network (CNN) architecture designed for image segmentation tasks. This architecture consists of an encoder and decoder. The encoder repeatedly applies convolutions and downsamples the image. It learns increasingly abstract features. The decoder upsamples these features back up to the original resolution of the input. Its job is to figure out the exact location of features. In the context of our project, encoded early layers will detect edges and features, while deeper layers can learn patterns associated with tumors. [?]

Each MRI scan consists of four images: T1, contrast-enhanced T1 (T1CE), T2, and T2-FLAIR. These images will be stacked as separate input channels, allowing the model to use information from each image concurrently in its prediction.

The target labels will consist of the already labeled segmentation masks provided with the dataset. Therefore, as the labels are pixel level, the model will also perform pixel-wise classification, assigning each pixel in the U-Net slice to either the background or one of the annotated tumor regions.

The model will be built based off a U-Net architecture, which is a tried and proven workflow designed specifically for biomedical image classification tasks. PyTorch will be used for implementation and training/validation.

4 TA Consultation

We consulted Yasmin Ahmadi regarding our proposed MRI segmentation project using the BraTS 2021 dataset. We discussed the available data, including the four MRI modalities (T1, T1CE, T2, and T2-FLAIR), and our plan to extract 2D axial slices from the 3D volumes for model training. We also discussed the target segmentation labels and our plan to use a U-Net architecture.

Yasmin recommended establishing a simpler CNN-based segmentation model as a baseline against which to evaluate U-Net. She suggested comparing models of increasing complexity to determine whether a more advanced model provides a meaningful improvement in segmentation performance, rather than evaluating U-Net in isolation. She also suggested considering binary segmentation depending on the final

definition of our target labels.

Based on this feedback, we plan to retain the four-class segmentation task defined by the BraTS annotations while introducing a simpler CNN-based segmentation model as our baseline. We will then implement U-Net and evaluate both models using the same patient-level training, validation, and testing splits. We plan to use the Dice similarity coefficient to measure overlap between predicted and ground-truth tumor regions. This will allow us to directly compare U-Net against the simpler CNN baseline and determine whether the additional architectural complexity produces a meaningful improvement in segmentation performance.

References

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. *CoRR*, abs/1505.04597, 2015.